

PRAKASH KUMAR

Patna, Bihar · +91 7324906288 · pr1624300@gmail.com · [LinkedIn](#) · [GitHub](#)

PROFESSIONAL SUMMARY

B.Tech student in Computer Science Engineering (AI & ML) with hands-on experience building production-grade AI applications, RAG pipelines, and full-stack web systems. Proficient in Python, React.js, and FastAPI, with strong expertise in LLM-based architectures, vector search, and retrieval-augmented generation. Skilled in end-to-end ML development, System design and cloud-native practices. Passionate about building intelligent, scalable systems that deliver measurable impact.

TECHNICAL SKILLS

Languages: C++, Python
ML / AI: Scikit-learn, LangChain, NumPy, Pandas, RAG Workflows, Pinecone, NLP
Web: React.js, Node.js, FastAPI, REST APIs
Tools: Git, GitHub, Docker
Databases: MySQL (SQL), MongoDB (NoSQL)

PROJECTS

Healthcare RAG Chatbot

Python | FastAPI | Streamlit | LangChain | Pinecone | Google Gemini Embeddings | Groq | MongoDB

- Designed and deployed a role-aware RAG-based healthcare assistant over PDF corpora; implemented recursive text chunking, Gemini embeddings, and Pinecone vector retrieval for accurate document-grounded responses.
- Integrated Groq LLM inference with source attribution, Basic Auth with bcrypt hashing, and role-based access control via FastAPI and MongoDB, restricting uploads to authorized users only.
- Optimized embedding flow with batching and dimension-safe vector configuration to align Gemini embeddings with Pinecone index requirements, reducing quota-related upload failures.

Intelligent Loan Approval System

Python | Scikit-learn | XGBoost | Pandas | FastAPI | React.js

- Built an end-to-end credit risk prediction system on 50K+ records across 5+ ML models, tuning XGBoost to 0.984 ROC-AUC and 92.6% accuracy via stratified cross-validation; automated testing with pytest and Selenium through GitHub Actions CI on every PR and push.
- Engineered financial risk features (DTI, LTI, PTI, net worth) with preprocessing pipelines for imputation, outlier capping, encoding, and scaling; containerized the backend with Docker and deployed via Render's image registry, eliminating environment mismatch failures.
- Delivered a production-ready FastAPI + React system with real-time single/batch inference, input validation, and explainable decision insights via a live dashboard; deployed the frontend on Vercel with automated CI/CD for zero-downtime updates.

Full-Stack Expense Tracker

React | Vite | Node.js | Express.js | MongoDB | Mongoose | Tailwind CSS | Framer Motion | JWT

- Built a full-stack finance tracker with real-time balance dashboard and category-wise spending analytics backed by per-user scoped REST APIs, JWT auth, and bcryptjs hashing.
- Implemented a tiered budget-alert system at 70%, 85%, and 95% of monthly income via warning toasts, dashboard cards, and a critical breakdown modal surfacing signals before limits are breached.
- Delivered a mobile-friendly UI with Framer Motion animations and Recharts visualizations on a React + Vite frontend backed by Node.js, Express, and MongoDB.

EDUCATION

B.Tech - Computer Science & Engineering (AI & ML)

2023 - Present

Sikkim Manipal Institute of Technology · CGPA: 8.02 / 10

Relevant Coursework: Machine Learning, Deep Learning, Database Management Systems, Operating Systems, Data Structures & Algorithms, Computer Networks

Class XII - CBSE · Kendriya Vidyalaya – 71.4%

2022

ACHIEVEMENTS

- 5-star Gold badge in Python on HackerRank; active coding club member focused on ML and web development.
- Self-directed learner: completed courses in deep learning, REST API design, and cloud-native deployment.